

SECURITY INCIDENT REPORT — CRITICAL

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INCIDENT SUMMARY

Field	Value
Incident ID	IR-2026-001
Date	June 11, 2026
Analyst	Houston — SOC Analyst
Severity	CRITICAL
Status	Under Investigation
Initial Alert	Suspicious Authentication from Kali
Alert Detection Time	21:20:43 EDT
Attack Window	19:43:50 — 19:52:15 EDT (approximately 9 minutes)
Domain Affected	hueylab.local
Domain Controller	WIN-430KUCVRBB4.hueylab.local

1. Executive Summary

On June 11, 2026, the Splunk SIEM alert 'Suspicious Authentication from Kali' fired at 21:20:43 EDT, indicating suspicious network authentication activity originating from a Linux workstation named KALI at IP address 192.168.99.20. Investigation revealed a successful Active Directory credential attack using CrackMapExec that compromised two domain accounts including the Administrator account, resulting in full domain compromise. This is a Critical severity incident requiring immediate containment.

2. Detection Details

2.1 Initial Alert

Field	Value
Alert Name	Suspicious Authentication from Kali
SIEM Platform	Splunk Enterprise
Detection Time	21:20:43 EDT
Trigger Condition	EventCode=4624, Workstation_Name=kali, Number of Results > 0

Field	Value
Alert Severity	High
Alert Type	Real-time

2.2 Splunk Investigation Queries

The following SPL queries were used during the investigation:

Query 1 — Initial triage:

```
index=main source="WinEventLog:Security" EventCode=4624 Workstation_Name=kali | table _time, Account_Name, Workstation_Name, Logon_Type, Authentication_Package
```

Query 2 — Scope analysis:

```
index=main source="WinEventLog:Security" EventCode=4624 Workstation_Name=kali | stats count by Account_Name | sort -count
```

Query 3 — Administrator logon timeline:

```
index=main source="WinEventLog:Security" EventCode=4624 Account_Name=Administrator | table _time, Workstation_Name, Logon_Type | sort _time
```

3. Attack Timeline

Time (EDT)	Event	Account	Source	Significance
19:43:50	Successful network logon	Administrator	KALI / 192.168.99.20	First Administrator compromise from Kali
19:49:16	Successful network logon	Administrator	KALI / 192.168.99.20	Second Administrator access
19:52:15	Successful network logon	Administrator	KALI / 192.168.99.20	Third Administrator access
21:20:43	Splunk alert fired	ksayian	KALI / 192.168.99.20	SIEM detection — investigation begins
21:20:43+	SOC investigation initiated	N/A	N/A	Analyst begins triage and scope analysis

4. Scope of Compromise

4.1 Compromised Accounts

Account	Logon Count	Privilege Level	Source Machine	Method
ksayian	2	Standard Domain User	KALI (192.168.99.20)	NTLM V2 / CrackMapExec
Administrator	3	DOMAIN ADMIN — Full Control	KALI (192.168.99.20)	NTLM V2 / CrackMapExec

4.2 Forensic Evidence

Indicator	Value
Windows Event ID	4624 (Successful Logon)
Logon Type	3 (Network Logon — Remote)
Authentication Package	NTLM V2
Logon Process	NtLmSsp
Source IP	192.168.99.20
Source Hostname	KALI
Target System	WIN-430KUCVRBB4.hueylab.local
Attack Tool Identified	CrackMapExec (SMB)

5. Root Cause Analysis

The attack succeeded due to the following security vulnerabilities:

Vulnerability	Severity	Description
Password Reuse	CRITICAL	The Administrator account used the same password as the ksayian domain user account. Once ksayian was compromised, Administrator access was trivially obtained.
No Account Lockout Policy	HIGH	No account lockout threshold was configured. CrackMapExec was able to authenticate repeatedly without triggering any lockout.
Weak Password Policy	MEDIUM	Minimum password length of 7 characters allowed simple passwords to be set across multiple accounts.
NTLM Authentication Enabled	MEDIUM	The server accepted NTLM authentication from external Linux machines, which is unusual and should be restricted.

6. Impact Assessment

- Full domain compromise — Administrator account accessed from unauthorized machine
- All domain resources potentially accessible including file shares, ADMIN\$ and C\$
- Domain controller WIN-430KUCVRBB4 potentially exposed to unauthorized commands
- All domain user credentials potentially at risk of extraction
- Active Directory structure potentially modified

7. Containment & Remediation Recommendations

Immediate Actions (Within 1 Hour)

- Isolate 192.168.99.20 (KALI) from the network immediately
- Reset ksayan password to a unique strong password
- Reset Administrator password to a unique strong password
- Review all actions performed during the 19:43-19:52 attack window
- Check for any new user accounts created or group membership changes

Short Term Actions (Within 24 Hours)

- Enable Account Lockout Policy — set threshold to 5 attempts, 30 minute lockout
- Increase minimum password length to 14 characters
- Enforce password uniqueness across all accounts
- Review and restrict NTLM authentication where possible
- Audit all privileged group memberships

Long Term Remediation

- Implement Privileged Access Workstations (PAW) for admin accounts
- Enable Multi-Factor Authentication for all privileged accounts
- Deploy additional Splunk alerts for Event ID 4625, 4648, and 4720
- Conduct quarterly Active Directory security reviews
- Consider deploying Microsoft Defender for Identity for advanced AD threat detection

8. Lessons Learned

- Password reuse across privileged and non-privileged accounts creates single points of failure
- The absence of an account lockout policy enables automated credential attacks
- Splunk real-time alerting successfully detected the attack — SIEM investment validated
- Forensic evidence in Windows Security logs provided complete attack reconstruction
- A single compromised standard user account led to full domain Administrator access

